Scratch: Loops, Events & Broadcasts

Make your sprites repeat, react and talk to each other!

Scratch | Beginner | Ages 6–12

The Big Idea

- An event block starts a script, like the green flag or a key press.
- A loop repeats blocks so you do not have to copy them again and again.
- A condition (if) lets your sprite decide what to do.
- Broadcasts let one sprite send a message that other sprites can hear.
- Clones let one sprite make many copies of itself while the game runs.

Events (Yellow Blocks)

- `when green flag clicked` starts the project.
- `when [space] key pressed` runs when you press that key.
- `when this sprite clicked` runs when you click the sprite.
- `when I receive [message]` runs when someone broadcasts that message.

Loops and Decisions (Gold Blocks)

- `repeat (10)` runs the blocks inside exactly 10 times.
- `forever` repeats until the project stops.
- `repeat until <condition>` repeats until something becomes true.
- `if <> then` runs once when true, `if / else` picks between two paths.
- `wait (1) seconds` pauses so things happen at a nice speed.

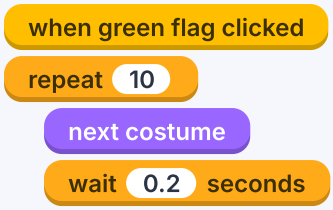
Broadcasts and Clones

- `broadcast [message]` sends a message to every sprite and the backdrop.
- `broadcast [message] and wait` waits until the other scripts finish.
- `create clone of [myself]` makes a copy of the sprite.
- `when I start as a clone` is the script that each copy runs.
- `delete this clone` removes a copy so the project does not slow down.

💻 Script Examples

Dance loop

Scratch blocks

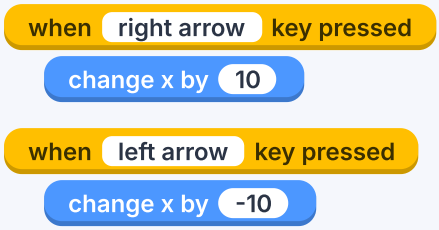


▶ What happens:

The sprite flips through its costumes 10 times, 0.2 seconds each, like a little dance.

Keyboard controls

Scratch blocks

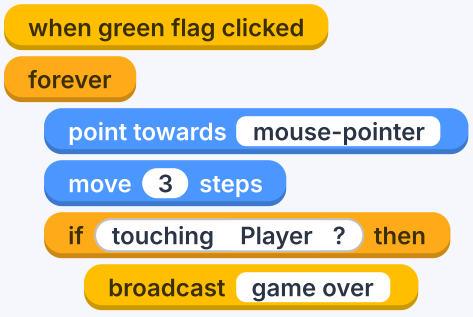


▶ What happens:

Pressing the right arrow moves the sprite right, and the left arrow moves it left.

A chaser (forever + if)

Scratch blocks



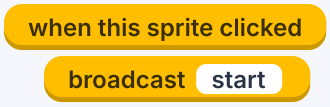
▶ What happens:

The chaser follows the mouse. If it touches the Player, it sends a game over message.

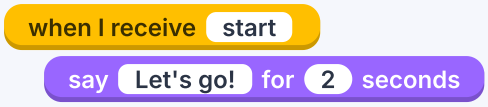
Broadcast between sprites

Scratch blocks

SPRITE 1:



SPRITE 2:



▶ What happens:

Clicking Sprite 1 sends start. Sprite 2 hears it and says Let's go!

Falling stars with clones

Scratch blocks



▶ What happens:

Ten stars appear one after another, each falls down the stage, and then it deletes itself.

📚 Key Terms

Event
Something that happens, like a click or a key press, that starts a script.

Script
A stack of blocks that run in order.

Loop
Blocks that repeat.

Forever
A loop that never stops by itself.

Condition
A true or false question, like touching a sprite.

Broadcast
A message sent from one sprite to others.

Clone
A temporary copy of a sprite.

Costume
One picture of a sprite. Changing costumes makes animation.

Sensing
Blocks that check things, like touching or key pressed.

Stop all
A block that ends every script in the project.

🧮 Quick Reference

`repeat (10)`	run 10 times
`forever`	run until stopped
`repeat until <touching [Edge]?>`	
`if <key [space] pressed?> then`	
`broadcast [start]`	send a message
`when I receive [start]`	listen for it
`create clone of [myself]`	make a copy
`delete this clone`	remove the copy

✏️ Worked Examples

Q: Make a sprite walk across the stage with 20 steps.

A: Step 1: when green flag clicked. Step 2: repeat (20). Step 3: Inside, add move (10) steps and wait (0.1) seconds. The sprite moves 200 steps in total.

Q: Make a Start button that begins the game.

A: Step 1: Make a button sprite. Step 2: when this sprite clicked, broadcast [start]. Step 3: On the player sprite, use when I receive [start] to begin the game script.

Q: Make a sprite jump when the space key is pressed.

A: Use when [space] key pressed, then repeat (10) with change y by (10), then repeat (10) with change y by (-10). The sprite goes up and comes back down.

🚫 Common Mistakes

- Putting blocks outside a hat block, so nothing starts them.
- Using `forever` when you wanted `repeat`, so the script never ends.
- Forgetting `wait`, so things happen too fast to see.
- Broadcasting a message and using a different name in `when I receive`.
- Making clones without `delete this clone`, which can slow the project down.

👨‍👩‍👧 Parent Corner

- Ask your child to tell you the story of the script: What starts it? What repeats?
- Play Simon Says: when you say a command, your child acts it out. That is an event.
- Challenge them to make a sprite that does something different on each key.
- If something does not run, ask which hat block should start it.
- Encourage a quick sketch of the idea on paper before they build it.

🎯 Quick Check

1. Which block repeats forever?

✅ The forever block.

2. Which block lets one sprite tell another sprite to act?

✅ broadcast [message], with when I receive [message] on the other sprite.

3. What starts a script when you press the space key?

✅ when [space] key pressed.

4. What does repeat (10) do?

✅ It runs the blocks inside it exactly 10 times.

5. What does delete this clone do?

✅ It removes the clone that is running the script.